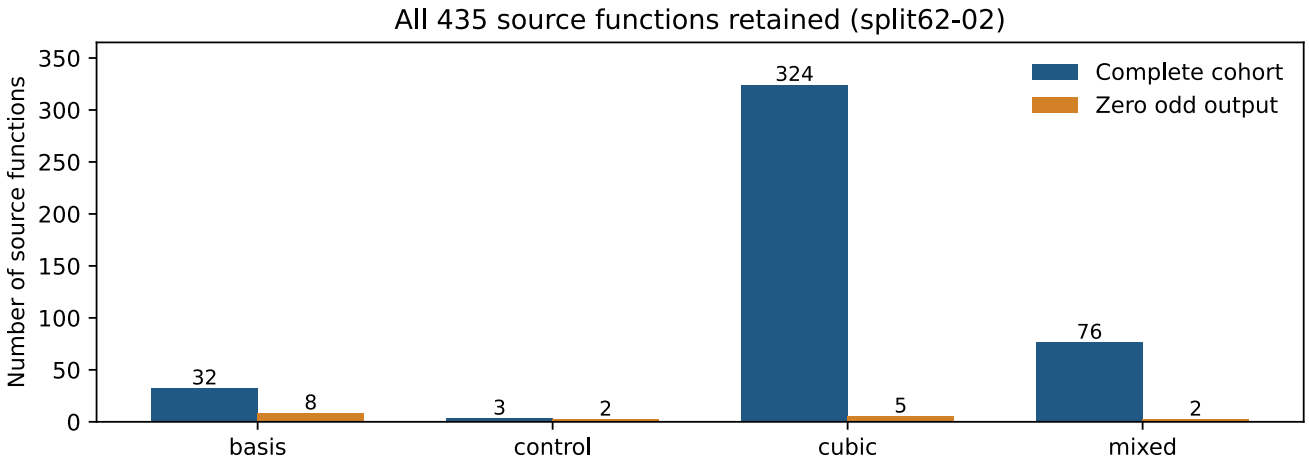
R62: complete square decomposition and generator route

Exact Koblitz modular-carrier field construction. Source field: the retained R61 degree32 extension of F2(u), with generators lambda and v. These are whole-source identities, with no scalar-fiber or series approximation.

Recorded computation

The matrix construction binds all 32 standard monomials and squared columns and checks both inverse products. The repaired complete cohort checks435 whole-source decompositions and13,920 rational coefficient parity identities. The fresh source replay reloads every input and root, binds all 435 functions to R61 and rechecks the complete operator and negative controls.

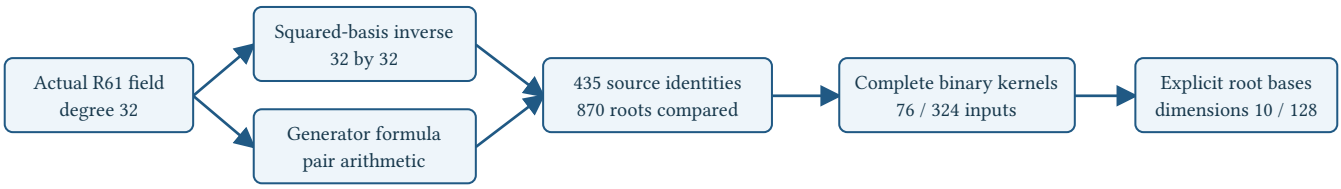


Recorded counts from split62-02; all 435 rows, seed62, one exact source instance. Zero odd components are field-operator outputs. Their complete supported divisors and global differential regularity are separate obligations.

Terminal receipts

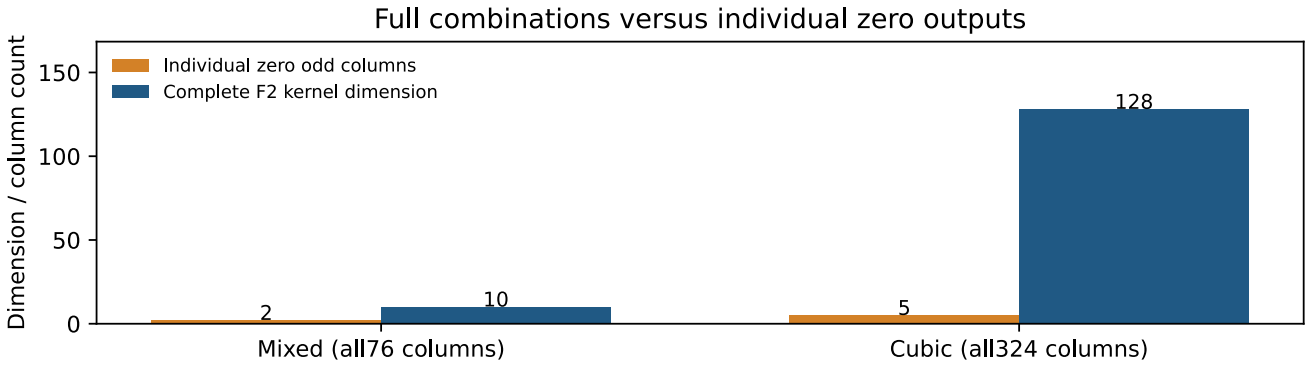
Attempt	Status	Native wall seconds
check62-01	accepted	68.0
direct62-01	accepted	87.0
kernel62-01	accepted	282.0
matrix62-01	accepted	4.0
rustcheck62-01	accepted	0.0
split62-01	rejected	84.0
split62-02	accepted	104.0
square-lean62-01	rejected	not recorded
square-lean62-02	rejected	not recorded
square-lean62-03	accepted	not recorded

Wall times are native receipt intervals, not controlled performance comparisons. Lean compile and audit intervals were not recorded by the proof driver.



Source identities, complete kernels and roots are separately checked stages.

Complete square subspaces and generator route



All76mixed and324 cubic functions are retained in the complete F2 kernel. Native checked denominator clearing and binary row reconstruction precede rank/nullspace computation; every resulting root is checked in the full source. The independent Rust checker replays both full binary matrices and kernels.

Direct generator route and proof scope

For a grouped characteristic-two source polynomial,

$$P(t) = a^2 + ub^2 + tc^2 + utd^2$$

. Put $q = c^2 + ud^2$. With $P(t) = 0$ and a checked nonzero derivative/inverse,

$$t = \left(\frac{ac + ubd}{q}\right)^2 + u\left(\frac{ad + bc}{q}\right)^2.$$

The direct route uses pair arithmetic rather than reading the squared-basis matrix. It checks the actual source grouping, derivative inverse, both source generators, all 32 generated basis pairs and all 435 functions, and compares 870 roots with the matrix output. The rational scalar parity helper is shared; this is not a separately implemented scalar parity algorithm.

Lean statements

The pinned Lean 4.34.1/mathlib d13f23b toolchain checks twelve authored claims: rational parity, uniqueness under nonsquare-u and characteristic-two premises, the zero-odd square criterion, five cyclic5 norm/projector/control claims, three pair arithmetic identities and the generator formula. Native field/model and nonsquare-parameter inputs, normality, local valuations, complete holomorphic regularity, actual deck/Cartier action and target-factor action are not inferred from these abstract implications.

Failures retained

The first complete 435 attempt used unsupported ideal equality and is rejected in full despite a late marker. The first Lean compile failed on a lemma name and numeral-cast normalization; the second lacked a Ring tactic import. Both artifacts are rejected. All later failures, if present, are listed in results.json with their actual errors. An early raw-index mistake was corrected without deleting local bytes or rewriting Git history. Accepted producers were not repeated for publication.

Scaling and next construction

GENERATOR-LEAD62.md and SCALE112-DESIGN62.md bind the proposed larger rung to the marked23 degree 7 base followed by the degree 16 lambda extension. The larger field has degree 112;991 denotes the selected factor action. Next: apply the generator operator to this exact larger source and complete the normal-order/infinity bridge needed for the full 61-dimensional holomorphic space. Unit divisors/saturation, actual primitive cyclic block, Hecke/order-11 action, proper polarization, quotient saturation and arbitrary reverse divisors retain their separate obligations.

Reproduction and provenance

PROTOCOL.md, protocol.json, donors.json and per-attempt source revisions freeze the affected inputs before data. results.json verifies historical source bytes. Native limits: 1200seconds, 4 GiB virtual, one Singular/FLINT thread, seed62, one math job. Lean: 1200seconds separately per compile/audit, CPU0, -j1 -M4096, unlimited virtual address space. No timing claim.

Sources: R61/SQUARE-DECOMPOSITION61.md; R52/CM.md; Singular manual.